O-D-825C
October 1, 1973
SUPERSEDING
Fed. Spec. O-D-825B
August 10, 1961

FEDERAL SPECIFICATION

DUPLICATING FLUID, DIRECT PROCESS (SPIRIT TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fluids for use in direct-process (spirit-type) duplicating operations (see 6.1).

1.2 Classification.

1.2.1 Types. Duplicating fluid shall be furnished in the following types, as specified (see 6.2):

Type I - Low methyl alcohol content.
Type II - High methyl alcohol content.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-228 - Paper Set, Duplicating Master Spirit or Liquid Process.
UU-P-232 - Paper, Duplicating, Copy, Liquid Process.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Standard No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7510

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

Laws and Regulations:

21 CFR, Part 191 - Food and Drug Administration Regulations under the Federal Hazardous Substances Labeling Act.

26 CFR, Part 212 - Internal Revenue Service (Publication No. 368, Revised); Formulas for Denatured Alcohol and Rum.

49 CFR - Department of Transportation Rules and Regulations for Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The documents listed below form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Duplicating fluid shall be as specified herein and shall be suitable for use with spirit-type duplicating machines and addressing machines.

3.2 Chemical composition. The chemical composition of duplicating fluid shall be as shown in table I (see 4.4.2).

3.2.1 Denatured alcohol content. The denatured alcohol content shall conform to Formula 3-A of Internal Revenue Service Formulas for Denatured Alcohol and Rum (see Note 1).

TABLE I. Chemical composition

Properties (per 100 gallons)	Type I	Type II
	Gallons	Gallons
Denatured alcohol, 190 proof (see 3.2.1)	95	35 to 55
Methyl alcohol	(see Note 1)	40 to 60
Ethylene glycol-monoethyl ether	5 (max.)	5

NOTE 1. Specially denatured alcohol Formula 3-A: Five gallons of commercially pure methyl alcohol is added to every 100 gallons of ethyl alcohol.
NOTE 2. The volumes shown above shall be based on the U.S. gallon at 15.5°C. (60°F.).

3.3 Performance. Duplicating fluid, when tested as described in 4.4.1, shall be capable of producing the number of copies specified in Federal Specification UU-P-228. All copies shall be legible with sharply defined characters of good intensity and without spotty appearance.

3.4 Appearance. The duplicating fluid shall be clear and shall be free from suspended material and sediment when tested as described in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

3.5 pH value. The pH value shall be not less than 5.5 nor more than 8.0 at 25°C. when tested as described in 4.4.4.

3.6 Stains or spots. Duplicating fluid shall not stain or spot when tested as described in 4.4.5.

3.7 Odor. The duplicating fluid shall not have nor give off, at any time during the duplicating operation, an objectionable odor and shall not leave a residual odor in the copy paper when tested as described in 4.4.6.

3.8 Copper corrosion. Duplicating fluid shall show no marked discoloration on copper when tested as described in 4.4.7.

3.9 Effect on rubber. Duplicating fluid shall not have any visible injurious effect on the rubber rollers or rubber parts of the duplicating machine when tested as described in 4.4.8.

3.10 Flash point. The flash point shall be not less than 60°F. (open cup) when tested as described in 4.4.9.

3.11 Workmanship. The duplicating fluid shall be manufactured in accordance with good commercial practice. The commodity shall be furnished free from evidence of dirt and any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL-STD-105, except where otherwise indicated herein.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified therein, or, if none, in accordance with this specification, or the contract or order.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects specified hereinafter and the acceptable quality levels (AQL's) specified in 4.2.2.1.5. The lot size for purpose of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of filled primary containers for examination in 4.2.2.1.1, 4.2.2.1.2 and 4.2.2.1.3 and in units of shipping containers for examination in 4.2.2.1.4.

4.2.2.1.1 Examination of the end item filled primary container for the characteristics and defects listed in table II. Examination of the end item filled container shall be made for closure, appearance, construction, workmanship, and markings. If one or more of the sample primary filled containers has one or more of the marking defects listed in table II as to the cautionary marking required in 5.4.3, the lot represented by the sample(s) shall be rejected.

TABLE II

Characteristics	Defects
Closure.	Any leakage or seepage of contents. Inner seals not provided when specified.
Appearance, construction, and workmanship.	Dented or crushed affecting serviceability. Indication of rust. Not in container specified. Not clean - any evidence of dirt, dust, or foreign matter. Nondrip pour spouts missing (when required). Exterior coating missing, or not as specified (when required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

TABLE II. (con.)

Characteristics	Defects
Markings (see 4.2.2.1.1 and 5.4.3).	Omitted, illegible, incorrect, incomplete, not in accordance with requirements. Precautionary marking printed on paper label attached to can. Precautionary marking incomplete, missing, illegible, not permanent on can. Marking smears; not clear. Not in accordance with MIL-STD-129 (for military).

4.2.2.1.2 Examination of the end item filled primary container for defects in net contents. The sample unit for this examination shall be one filled primary container. The average net contents of the sample units shall be not less than specified or indicated quantity.

4.2.2.1.3 Examination of the end item filled primary container for characteristics and defects listed in table III. The sample unit for this examination shall be one filled primary container. Examination of the end item filled primary container shall be made for appearance and odor of duplicating fluid for the defects listed in table III.

TABLE III.

Characteristics	Defects
Appearance of duplicating fluid.	Presence of dirt and any foreign matter, suspended material, or sediment. Not clear liquid. Not homogeneous.
Odor.	Objectionable odor of fluid.

4.2.2.1.4 Examination for defects in packaging, packing, and markings. An examination shall be made to determine compliance with the requirements of Section 5 (see table IV). This examination is not applicable when the primary container is the shipping container.

TABLE IV.

Characteristics	Defects
Packaging (as applicable).	Not in accordance with contract requirements. (See table II).
Packing (as applicable).	Unit cans not packed as specified. Unit cans do not fit snugly into container. Number of cans less than specified or indicated quantity. Not in accordance with contract requirements. Container not as specified; closures not accomplished by specified or required methods or materials.
Weight.	Gross or net weight exceeds specified requirements.
Markings.	Omitted, illegible, incorrect, incomplete, not in accordance with contract requirements (see 5.4.3).

4.2.2.1.5 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels and the acceptable quality levels (AQL's) expressed in defects per 100 units, shall be as follows:

	Inspection level	AQL's
For examination in 4.2.2.1.1	I	1.5, except as otherwise specified in 4.2.2.1.1.
For examination in 4.2.2.1.2	L-4	------
For examination in 4.2.2.1.3	L-4	2.5
For examination in 4.2.2.1.4	L-4	4.0

4.3 Testing of the end item. The end item shall be tested for the applicable characteristics as specified in table V for each lot presented for inspection. The sample unit shall be one quart of duplicating fluid. The sample size shall be five. Each sample unit shall be taken from a different shipping container of the lot. The test sample shall consist of a two-quart composite of five sample units. Duplicate determinations shall be performed on the composite unless otherwise specified. There shall be no evidence of failure to meet the specified requirements. The requirement for ethylene glycol-monoethyl ether in the chemical composition of the duplicating fluid may be satisfied by acceptance of a certificate of compliance from the supplier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

TABLE V. End item testing requirements

Characteristic	Specification reference		Number of determinations [1]	Results reported as	
	Requirement	Test method		Pass or fail [2]	Numerically to nearest
Performance	3.2	4.4.1	1 on composite	X	---
Methyl alcohol content	3.3 (table I)	4.4.2	Average of 2 on composite	----	0.1%
Appearance	3.4	4.4.3	2 on composite	X	---
pH value	3.5	4.4.4	Average of 2 on composite	----	0.1 pH
Stains or spots	3.6	4.4.5	2 on composite	X	---
Odor (residual)	3.7	4.4.6	1 on composite	X	---
Copper corrosion	3.8	4.4.7	2 on composite	X	---
Effect on rubber	3.9	4.4.8	2 on composite	X	---
Flash point (open cup)	3.10	4.4.9	2 on composite	----	0.1°F.

[1] Test reports shall include all values on which results are based.

[2] If failure is indicated, report descriptions of failure.

4.4 Tests. Duplicating fluid selected as specified in 4.3 shall be subjected to the following tests. If the specimen fails one or more of these tests, the lot represented by the sample shall be rejected.

4.4.1 Performance. Duplicating fluid shall be tested in accordance with the procedures specified in UU-P-228 using class 2 grade A master sets and copy paper conforming to the requirements of UU-P-232. Copies shall be examined for the quality of reproduction specified in 3.2.

4.4.2 Methyl alcohol content. The methyl alcohol content of the duplicating fluid shall be determined by the following method (see 6.3).

4.4.2.1 Reagents.

a. Potassium permanganate solution. Dissolve 3.0 grams potassium permanganate and 15.0 milliliters of o-phosphoric acid in 100 milliliters water. Renew every 4 weeks.

b. Sodium salt of chromotropic acid (sodium 1.8-dihydroxynaphthalene-3,6-disulfonate) - 5 percent solution. Filter if not clear. Renew every week. The reagent may be purified by the following procedure:

Dissolve 10 grams chromotropic acid or its salt in 25 milliliters water. (Add 2 milliliters sulfuric acid to aqueous solution of the salt to convert it to free acid.) Add 50 milliliters of methanol, heat just to boiling and filter. Add 100 milliliters isopropyl alcohol to precipitate free chromotropic acid. (Add more isopropyl alcohol to increase yield of purified acid.)

4.4.2.2 Preparation of sample. Dilute or adjust the sample to a total alcoholic concentration of 5 to 6 percent. Using a 50-milliliter sample, distill through a simple still, collecting 40 milliliters distillate. Dilute to 50 milliliters with distilled water. (If alcohol has been determined previously, distillate may be adjusted to 5 to 6 percent alcohol concentration and used for this test.) If more than 0.05 percent methanol by volume is present, dilute to approximately that concentration with 5.5 percent alcohol. For samples containing less than 0.05 percent methanol, measure 200 milliliters into efficient fractionating still, place system under total reflux 15 minutes, and then slowly distill at high rate of reflux (at least 20:1). Collect 10 milliliters distillate and dilute to 160 milliliters with water.

4.4.2.3 Determination. Pipet 2 milliliters potassium permanganate solution into a 50-millilter volume flask. Chill in ice bath, add 1 milliliter chilled diluted sample, and allow to stand 30 minutes in ice bath. Decolorize with little dry sodium bisulfite and add 1 milliliter chromotropic acid solution. Add 15 milliliters sulfuric acid slowly with swirling and place solution in hot-water bath (60 - 75 C.) for 15 minutes. Cool, add enough water to dilute approximately to 50-milliliter mark, mix, and bring to volume with water at room temperature. Read absorbance at 575 mμ, using reagent blank of 5.5 percent alcohol treated similarly as reference. Treat a standard methanol solution containing 0.025 percent by volume methanol in 5.5 percent alcohol simultaneously in same manner, and read absorbance. (Temperature of standard and sample should agree within 1 degree since temperature affects depth of color.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

Calculate quantity of methanol in sample as follows:

Percent methanol equals $(A_S \div A_m) \times 0.025 \times F$, where A_S equals absorbance of sample, A_m equals absorbance of standard methanol, and F equals dilution factor of sample.

Example: Sample was diluted 25 times; absorbance of sample equals 0.421; absorbance of standard methanol equals 0.368. Then $(0.421 \div 0.368) \times 0.025 \times 25$ equals 0.715 percent methanol.

(If color of sample is too intense, dilute with sulfuric acid-alcohol blank prepared as above. No more than 3-fold dilution is permitted, as ratio of chromotropic acid to formaldehyde is too low if dilution is greater.)

4.4.3 Appearance. Pour some of thoroughly mixed sample into a test tube. Allow to stand in an upright position, at room temperature (20° to 30°C.) undisturbed for 10 minutes. Examine by transmitted light for clarity, uniformity, and freedom from suspended matter and sediment.

4.4.4 pH value. Dilute a portion of the sample with an equal volume of freshly boiled and cooled distilled water; mix thoroughly and determine the pH value at 25°C. with an electric pH meter having a glass electrode.

4.4.5 Stains or spots. Place 5 drops of the sample in the center of a quantitative filter paper, so supported that the center of the paper is not in contact with any object. Allow to evaporate, away from direct sunlight, at 20° to 30°C. Examine for stains or spots after a 2-hour drying period.

4.4.6 Odor. The odor of the duplicating fluid shall be determined in the original container. The odor shall again be determined during the performance test while the machine is in operation. Residual odor shall be determined in the pack after allowing the copy paper, from the performance test, to dry for 3 hours in a well-ventilated room at 20° to 30°C.

4.4.7 Copper corrosion. Place a clean strip of mechanically polished sheet copper, 1/2 inch wide and 3 inches long, in a glass tube 3/4 inch in diameter and 18 inches long, closed at one end. Add sufficient duplicating fluid to cover the copper strip. Heat the assembly to near boiling for a period of 30 minutes and examine the copper strip for discoloration. Marked blackening or irridescence (a slight tarnish shall be disregarded) shall be considered as not conforming to the requirements of this specification.

4.4.8 Effect on rubber. Place a rubber roller from a duplicating machine in a glass-stoppered container and cover the roller with the duplicating fluid. Allow to stand 24 hours. The duplicating fluid shall not produce any swelling or other deleterious effects on the roller.

4.4.9 Flash point. The flash point of the duplicating fluid shall be determined as described in Fed. Test Method Standard No. 791, method 1103.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2). When specified (see 6.2), nondrip pour spouts shall be furnished.

5.1.1 Level A. Duplicating fluid shall be packaged in a 1-gallon capacity rectangular can conforming to PPP-C-96, type V, class 4 or 5. Inner seals shall be provided when class 4 cans are used. Exterior coating shall be plan A or B, as specified (see 6.2).

5.1.2 Level B. Duplicating fluid shall be packaged as specified in 5.1.1, except that exterior coating shall be plan A. If required by the nature of the product, the inside of the can shall be completely coated with an enamel which will neither adversely affect, nor be adversely affected by, the packaged product.

5.1.3 Level C. One gallon of duplicating fluid shall be packaged in accordance with the manufacturer's standard practice.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Duplicating fluid, packaged as specified in 5.1, shall be packed in accordance with the level A requirements of the Appendix to PPP-C-96.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Six (6) gallons of duplicating fluid, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, type CF, variety DW, grade 275, style RSC. The box shall be closed in accordance with the appendix to the box specification. When specified (see 6.2), the duplicating fluid shall be packed in a close-fitting box conforming to PPP-B-636, grade V15c, style RSC.

5.2.3 Level C. Duplicating fluid, packaged as specified in 5.1.3, shall be packed in containers that will assure carrier acceptance and safe arrival at destination in accordance with the Uniform Freight Classification or National Motor Freight Classification. Shipping containers shall comply with the U.S. Department of Transportation requirements as specified in Title 49 CFR, parts 100-199.

5.3 Marking.

5.3.1 Civil agencies. In addition to the marking specified in 5.3.3 and any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123. The packages and shipping containers shall also be marked with the expiration date and shelf life markings as specified in Fed. Std. No. 123.

5.3.2 Military activities. In addition to the marking specified in 5.3.3 and any special markings required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

5.3.3 Precautionary marking. Each can shall be prominently and permanently marked with applicable Department of Transportation (D.O.T.) Regulations as shown below.

CAUTION: FLAMMABLE MIXTURE
DO NOT USE NEAR FIRE OR FLAME

POISON - DO NOT USE INTERNALLY (SKULL AND CROSSBONES SYMBOL)

USE ONLY IN WELL-VENTILATED WORK AREA (TO AVOID TOXIC AND CUMULATIVE EFFECTS)

CAUTION, AVOID INHALATION OF FUMES

ANTIDOTE: Give an emetic immediately so as to produce vomiting, or use a stomach pump. Follow by repeated doses of bicarbonate of soda (2 teaspoonfuls in water) or white of egg.

6. NOTES

6.1 Intended use. The duplicating fluid covered herein is intended for use with duplicating machines and addressing machines, spirit-process. Type I duplicating fluid has less health hazard than type II. Type II duplicating fluid will produce copies more intense in color than type I, but due to the higher methyl alcohol content permitted (see table I), presents a health hazard unless used under conditions of ADEQUATE VENTILATION. (See 5.3.3.)

6.2 Ordering data. Purchasers should exercise any desired options offered herein (see 5.1) and should specify the following (see 4.1).

(a) Title, number, and date of this specification.
(b) Type of duplicating fluid required (see 1.2.1).
(c) Data, as required, regarding examination and tests; and inspection responsibility for civil agency procurement if other than as specified (see 4.1).
(d) Whether pour spouts are required (see 5.1).
(e) Level of packaging and level of packing and any special marking required (see 5.1, 5.2 and 5.3).
(f) Exterior coating plan required for Level A packaging (see 5.1.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.

6.3 Methanol determination. The test method specified in 4.4.2 for the determination of methanol is in conformity with the Official Method of Analysis of the Association of Official Agricultural Chemists, 9th Edition, 1960.

DOD Coordination has been waived.

User interest:

Army - GL

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:15Z
Check the source to verify that this is the current version before use.